A model for ant food collecting behavior in a dynamical environment

Alex Goldhoorn

Wouter Lieffijn

University of Groningen, The Netherlands

Faculty of Behavioral and Social Sciences

Department of Artificial Intelligence

goldhoorn@ai.rug.nl / student number 1434284

w.lieffijn@ai.rug.nl / student number 1277154

1 Introduction

To inform each other over a food source, ants mark their paths to the food source using pheromones. Because an ant leaves a pheromone trail when it goes back from the food source to the nest, other ants can use that same path to find the way to the food source.

1.1 NetLogo-model

There is a NetLogo model where artificial ants use pheromones to 'inform' each other about a found food source in the vicinity of their nest. The model is a world of 35 × 35 cells with a nest in the centre, three food sources and 125 ants.

The ants move randomly one step forward in de direction of the Moore neighborhood, until they find a patch with food. They take a part of the food and return to the nest. They return directly to the nest, thereby assuming an ant knows the direction to its nest. While it returns it leaves pheromone 'drops' every step.

Every time step some pheromones evaporate and diffuse over the Moore neighborhood of the original patch. These two processes make the pheromones a kind of dynamical memory. The paths spread due to diffusion and fade because the evaporation.

The spreading of the pheromones creates a certain margin in the search space. When the pheromones would not diffuse, the ant should have to be in the exact patch to recognize the signal. Also the ant has to stay exactly on the path to prevent losing it. It allows the ants to get off the exact path and search for food.

The NetLogo model clearly shows that when there are no pheromone paths yet, the ants move randomly. But as soon as paths arise, an increasing number of ants follow the paths.

2 Our model

The NetLogo model is limited because of the way it processes the created program. This makes NetLogo slow, especially with large fields. Therefore the possibilities to extend the model and make it dynamical are rather limited. Therefore we chose to write an extended model in Java. Although NetLogo itself is a Java program, it has to interpret the written program and this makes it slower. This step is avoided when programming directly in Java.

The field size was increased to 128 × 96 cells; this is a surface ten times of the NetLogo model. The distribution of the food is also different. In the NetLogo model the food sources were placed at three fixed points and they contained a limited amount of food. When the food sources were finished there was no food and the ants could only walk randomly. In our model the food grows semi-randomly. At places where there already is food in the neighborhood, the chance of new food to grow is higher. This creates patches of food. To prevent the whole field to be covered with food a maximum of food in the 'world' was added.

To minimize computational power we chose to use a Von Neumann neighborhood for the ants for the movement and sensation. Because of this the movement and sensation can be executed by using simple computations. A drawback is that the movements are less smooth. Ants can move by doing one step forward or to turn to the left or right. Also the sensations of the ants are less accurate due to the use of a Von Neumann neighborhood. Ants can only sense the cells in front, to the left and right.

3 **Results**

In this section we discuss the results of one simulation of about 140,000 time steps. For the variables used during the simulation, see Attachment: variables used.

'Attachment: screenshots' shows three screenshots of a run of the simulation. It can be seen that at the start simulation (figure 1a) the food is distributed over the entire field. The ants seem to get the food mainly from the closest food piles. From time step 10,500 (figure 1b) the food is mainly growing at the borders of the field. The same effect is showed by figure 2. The graph shows the distance from the nest to the spot a food is picked up. Until time step 2,000 there is a clear increase in distance. This is due to the fact that the food patches near the nest are gone. The remaining time the distance fluctuates.

The effect of food patches growing only near the borders can be seen as an emergent effect. It is a result of the way the ants forage their food and the way the food grows in the simulation. The food has a higher probability to grow near other food patches. And because ants remove food near the nest faster, the patches near the nest have less time to grow back.

The number of food piles can also be roughly estimated by using a computer vision algorithm named connected components. The result can be seen in figure 3. The number of food piles seems to grow until about time step 20,000.

The mean time between two food pickups (not necessarily the same ants) is 14.3 (with a standard deviation of 21.5). This value does not change significantly over time.

Finally we can look at the scent (pheromones) and its distribution, see figure 4. The mean scent per patch of the simulation is 0.056 and its standard deviation is 0.223. The higher the mean scent, the more scent there is in the field. And the lower the standard deviation the more equally distributed the scent is. When the standard deviation is high, there are some patches with scent and some without. This is the case when there are 'paths' of scent to the food.

From figure 4 can be seen that sometimes there is more scent and the standard deviation is high, this can be seen in the simulation when there are a few strong paths as in figure 1a and b. And sometimes there is less scent as in figure 1c. These increases and decreases can be explained by the evaporation rate of the scent and the way the ants look for their food. When an ant finds food, it leaves a trail of scent while it goes to the nest. Other ants follow that trail to the food source and thereby increasing a path and the amount of scent (and the standard deviation of the scent per patch). When all the food in that pile is gone the ants stay near the path for a while because the scent is not yet completely evaporated, but they do not drop scent anymore. Therefore the mean scent per patch and standard deviation both decrease. When all the scent of that path is evaporated all the ants will again search randomly, another ant may find food and the same event repeats itself.

4 **Conclusion**

Although the original NetLogo model showed how a self-organizing colony of ants may find its food sources, it was a very simple model, primarily because of the static environment. We tried to extend the model with a dynamical environment in which the position of food sources changes over time.

A drawback of the model was that of using the Von Neumann neighborhood instead of the Moore neighborhood for perception and motion. However it seems that the colony of artificial ants does not have any problem with that.

Although it may happen that when some food source exhausts, the ants using the pheromone trail to it, are trapped by searching to the highest concentration of pheromones, it seems that these artificial ants can handle the dynamical environment very well.

5 Attachment: variables used

Field size	128 × 96 patches
Nest location	(64, 48)
Number of ants	100
Maximum food in field	5% of 128 × 96 = 614
Maximum food ant carries	100
Evaporation of pheromones	0.01
Diffusion of pheromones	0.1
Upper [pheromone] boundary	0.20
Lower [pheromone] boundary	0.0075
Probability of following pheromone trail	0.4
Probability of random turn	0.15

6 **Attachment: screenshots**

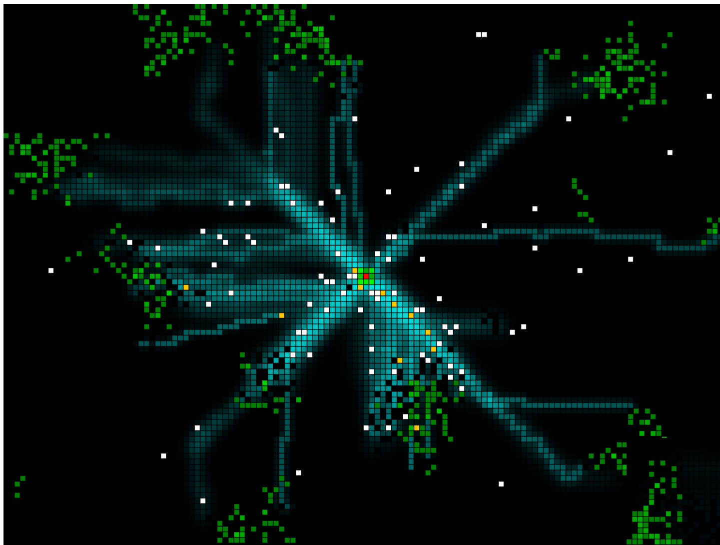


Figure a: time step 350

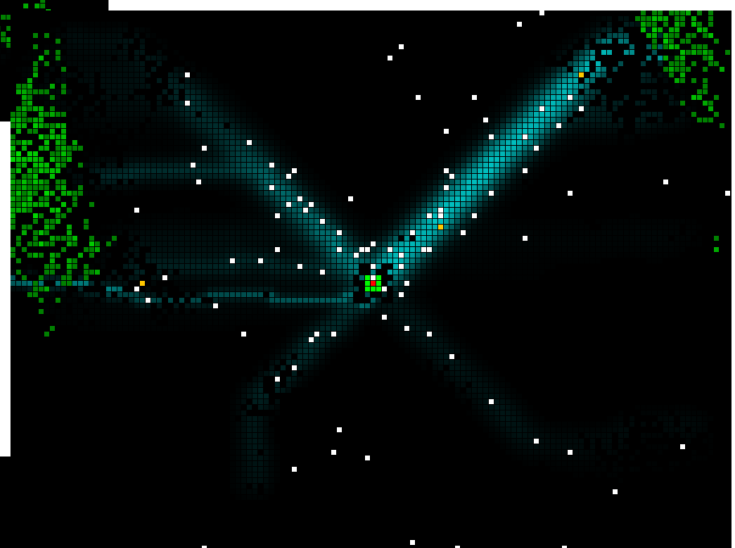


Figure b: time step 10,500

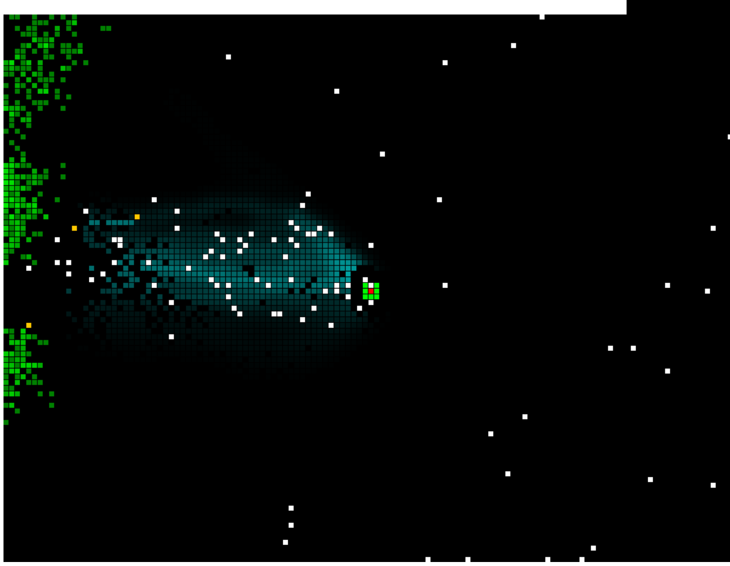
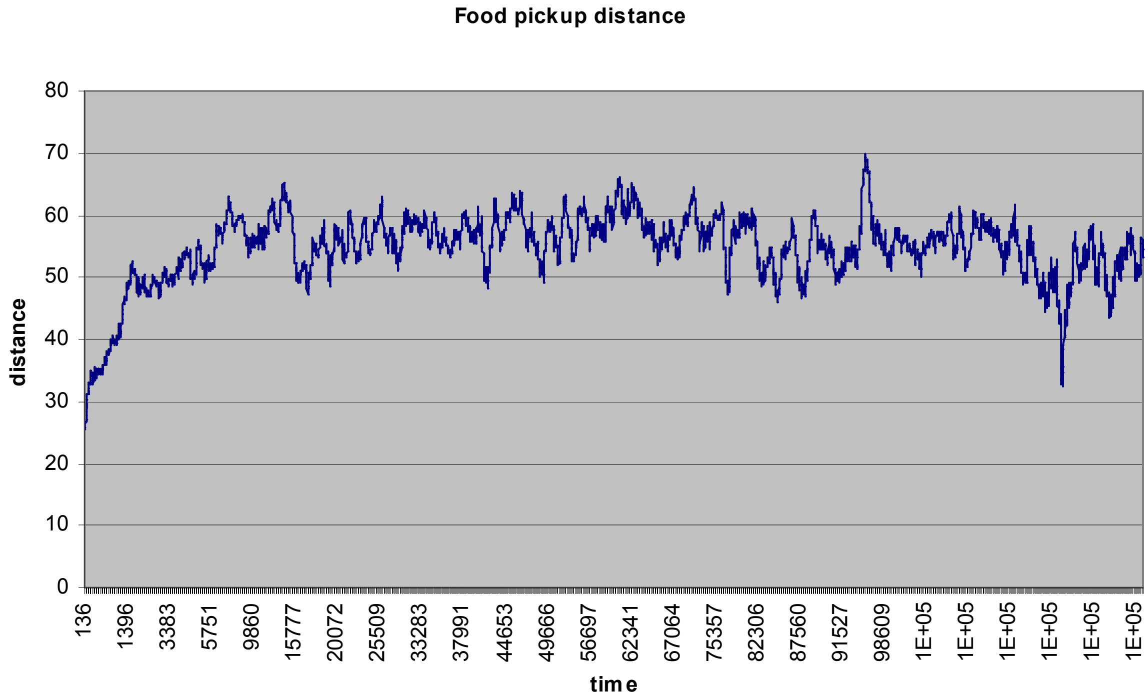


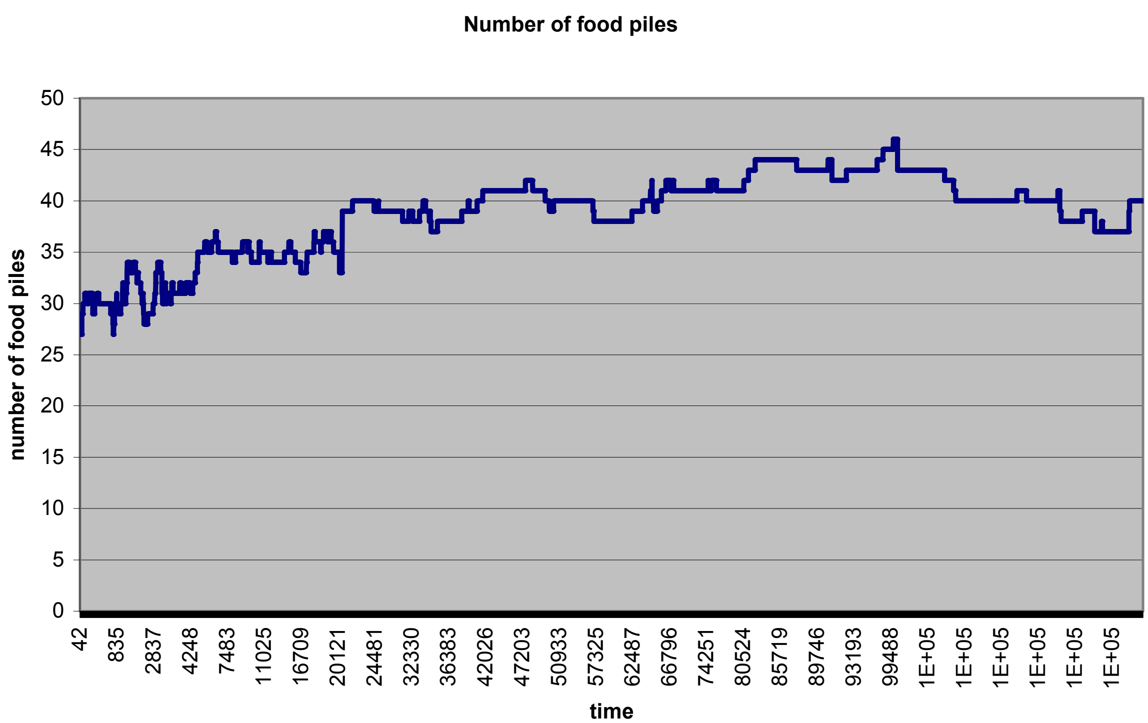
Figure c: time step 105,000

7 **Attachment: Food pickup distance**



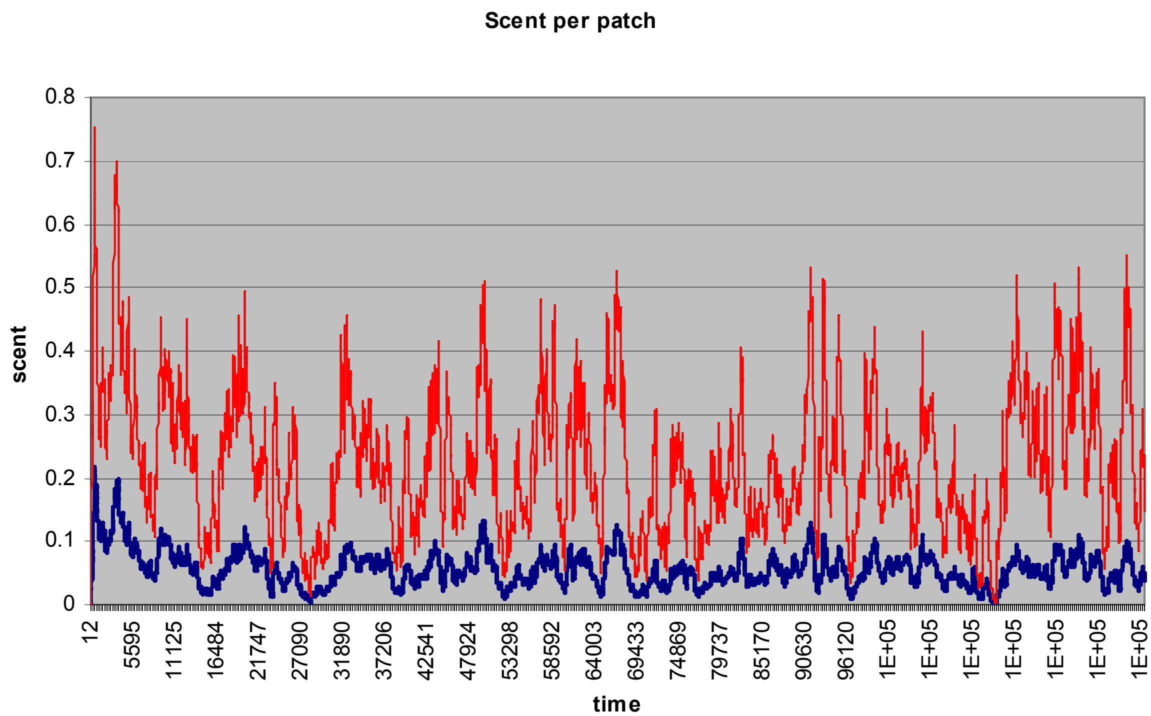
The distance to the nest from which an ant picks up a food patch. The graph is smoothed. The mean value is 54.6 and the standard deviation is 19.2.

8 **Attachment: number of foodpiles**



Number of food piles. The mean value is 38.7 and the standard deviation is 3.8.

9 **Attachment:**



Mean (blue series) and standard deviation (red series) of scent per patch over time